

Volume of revolution

Key Points

The volume of revolution for a function $y(x)$ is

- $\int_a^b \pi y^2 dx$ around x -axis between a, b
- $\int_c^d \pi x^2 dy$ around y -axis between c, d

These formula's are derived by taking small slices of the volume which are cylinders hence $\delta V = \pi r^2 \delta l$ where δl is the small length/height of the cylinder

Basic Skills

- Q1.** Name a function which when revolved around the x -axis forms a cylinder
- Q2.** For the function $y = 2x$
- a) Sketch the function
 - b) If the function is revolved around the x -axis between 0 and 5 what shape is formed
 - c) Find the volume of revolution around the x -axis between 0 and 5
 - d) Using the formula for the volume of the shape check your answer to part c)
 - e) Determine if the volume of revolution around the y -axis between 0 and 5 is the same

Competency Skills

- Q3.** For the function $y = 2x + 1$ name the shape and find the volume of revolution when a revolution around the x -axis between 0 and 5 is performed
- Q4.** Calculate the volume of revolution around the x -axis for $y = 1 - e^{-x}$ between $x = 0, \ln 2$
- Q5.** A model for a silhouette of a vase has $y = 4 - 2x + x^3$ for $-2 \leq x \leq 0$, with the units in x being 10cm find the volume of the vase by performing a revolution around the x -axis
- Q6.** Given that the volume of revolution around y -axis for $y = x^a$ between $y = 0, 1$ is 1 find a
- Q7.** A region R is formed by the enclosed area between $y = x^2 - 1, x = 0, y = 0, y = h$ for some height h
- a) When $h = 8$ find the volume of the enclosed region when it is revolved around the y -axis
 - b) Find the relationship between two heights such that the volume is in the ratio 1:2

Advanced Skills

- Q8.** By using $y^2 = |x|$ and the volume of revolution find the volume of an hourglass shape with height h and final radii r
- Q9.** For the reciprocal function $\frac{1}{x}$ for $a \leq x \leq b$ for some constant a
- a) Determine if the volume of revolution around the x -axis exists for all $a, b \in \mathbb{R}$
 - b) Let $a = 1$ and determine if the volume of revolution around the x -axis exists as $b \rightarrow \infty$
 - c) It is suggested that "The volume of a shape of infinite length is always infinite", comment on the statement
- Q10.** Derive the formula $\int_a^b \pi y^2 dx$

Extension

Q11. Prove that the radius of a torus (doughnut) of inner tube radius r and centre line from centre radius R is $2\pi^2 r^2 R$

Hint: Revolve the semi-circle $y = \sqrt{r^2 - x^2} + R$ for $-r \leq x \leq r$